



DESCRIPTION OF COURSEWORK

Course Code	MAT314
Course Name	Regression Analysis
Lecturer	Lim Min
Academic Session	2025/04
Assessment Title	Project 1: Simple Linear Regression

A. Introduction

In this project, you will use a simple linear regression model to predict used car price of a certain brand based on its mileage driven.

B. Course Learning Outcomes (CLO) covered

At the end of this assessment, students are able to:

CLO 3 Perform regression analysis using the statistical software R.

C. University Policy on Academic Misconduct

1. Academic misconduct is a serious offense in Xiamen University Malaysia. It can be defined as any of the following:
 - i. **Plagiarism** is submitting or presenting someone else's work, words, ideas, data or information as your own intentionally or unintentionally. This includes incorporating published and unpublished material, whether in manuscript, printed or electronic form into your work without acknowledging the source (the person and the work).
 - ii. **Collusion** is two or more people collaborating on a piece of work (in part or whole) which is intended to be wholly individual and passed it off as own individual work.
 - iii. **Cheating** is an act of dishonesty or fraud in order to gain an unfair advantage in an assessment. This includes using or attempting to use, or assisting another to use materials that are prohibited or inappropriate, commissioning work from a third party, falsifying data, or breaching any examination rules.
2. All assessments submitted must be the student's own work, without any materials generated by AI tools, including direct copying and pasting of text or paraphrasing. Any form of academic misconduct, including using prohibited materials or inappropriate assistance, is a serious offense and will result in a zero mark for the entire

assessment or part of it. If there is more than one guilty party, such as in case of collusion, all parties involved will receive the same penalty.

D. Instructions to Students

This is an individual project. In this project, you are required to perform a complete regression analysis using a simple linear regression model on the data set given to you.

The analysis must be done using the statistical software R.

The report must be typed using L^AT_EX. A template is provided on Moodle.

You are required to submit 2 files:

1. A .txt file that consists of your R code: **ABC1234567_P1_Rcode.txt**
2. A .pdf file that consists of your report: **ABC1234567_P1_Report.pdf**

E. Evaluation Breakdown

No.	Component Title	Marks
1.	Report	100
	TOTAL	100

F. Tasks

- Data Source

1. Download the data file from Moodle. The file name is your student ID. For example **MAT2309001.txt**.
2. Open R and change the working directory to where you saved the data file.
3. Read the data file into R.

```
> CarData <- read.table("MAT2309001.txt", header = TRUE)
```

4. Split your data into two sets: estimation data (training data, 70%) and prediction data (test data, 30%). Use the **set.seed()** function in R so that the results can be reproduced. For example,

```
> set.seed(1234)
> dt <- sort(sample(nrow(CarData), nrow(CarData)*.7))
> CarData_e <- CarData[dt, ]
> CarData_p <- CarData[-dt, ]
```

- Data Analysis

1. Produce a scatter diagram for **mileage** and **price** using the estimation data. Label the axes appropriately. No main title is needed, you will name the figure in L^AT_EX. This is Figure 1 in your report.
2. Produce a scatter diagram for **mileage** and **log(price)** using the estimation data. Label the axes appropriately. No main title is needed, you will name the figure in L^AT_EX. This is Figure 2 in your report.

3. Compute the sample correlation coefficient between `mileage` and `log(price)` using the estimation data.
 4. Fit a simple linear regression model to the estimation data with `log(price)` as the response and `mileage` as the regressor. Print the summary output.
 5. Use the fitted function obtained in 4 to predict `price` (store them in `price.p`) for values of `mileage` in the prediction data. Note that $e^{\log(y)} = y$.
 6. Produce a scatter diagram for `price` from the prediction data and `price.p`. Label the axes appropriately. No main title is needed, you will name the figure in L^AT_EX. This is Figure 3 in your report.
 7. Compute a 95% confidence interval for the mean `price` of used cars that have been driven 9,446 miles using the fitted function obtained in 4.
 8. Compute a 95% confidence interval for the mean `price` of used cars that have been driven 18,513 miles using the fitted function obtained in 4.
 9. Compute a 95% prediction interval for the `price` of a used car that has been driven 9,446 miles using the fitted function obtained in 4.
 10. Compute a 95% prediction interval for the `price` of a used car that has been driven 18,513 miles using the fitted function obtained in 4.
- Report Writing
 1. Download the L^AT_EX template from Moodle.
 2. Open the L^AT_EX template.
 3. Fill in the Submission Due Date in the format of dd/mm/yyyy.
 4. Fill in your Student ID and Student Name.
 5. Fill in the Date Received in the format of dd/mm/yyyy. “Date Received” is the date you submit the project. You may fill in this at a later time. But, do not forget to do it!
 6. Sign and date on page 3. Upload your signature to replace the current one. Sign, not type. Check if your signature is in the right size. The date on this page should be the same as “Date Received”.
 7. Section 1: Introduction

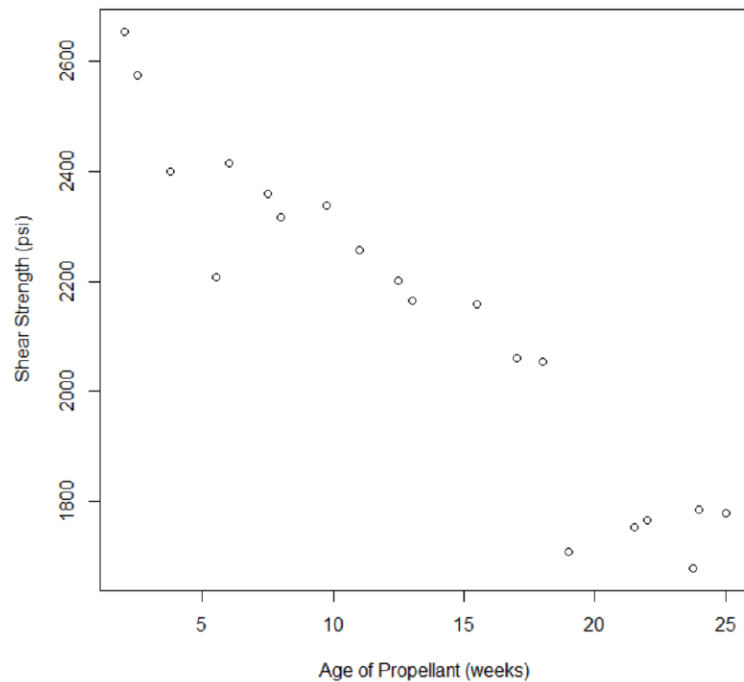
In this section, provide information of your data set (sample size, description of the variables used in the study, estimation data, and prediction data) and describe the purpose of the analysis (a summary of the 10 tasks under Data Analysis).

For your information, variables in the original data set are model, registration year, price (Euros), transmission type of gear box (auto, manual, others), distance driven (miles), fuel type (petrol, hybrid, others), road tax (Euros), miles per gallon, and engine size (litres) for each used car of a certain brand.
 8. Section 2: Data Analysis

Present **ALL** results from the 10 tasks under Data Analysis here. No R code should be shown in this section.

Figures can be attached but output must be retyped.

Use `includegraphics` to attach figures. Figures should not come with an outer boarder. Avoid stretching the figures in either direction. Ensure the labels are large enough to read. An ideal figure is provided below.



Output can be reproduced in a tabular form. An example is shown below. Note that $1.64\text{e-}10$ needs to be retyped as 1.64×10^{-10} . And, not all parts are reproduced.

R output:

```
> summary(prop.model)

Call:
lm(formula = strength ~ age)

Residuals:
    Min       1Q   Median       3Q      Max
-215.98  -50.68   28.74   66.61  106.76

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  2627.822     44.184   59.48  < 2e-16 ***
age          -37.154      2.889  -12.86 1.64e-10 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 96.11 on 18 degrees of freedom
Multiple R-squared:  0.9018,    Adjusted R-squared:  0.8964
F-statistic: 165.4 on 1 and 18 DF,  p-value: 1.643e-10
```

Reproduced table:

	Estimate	Std. Error	<i>t</i>	<i>P</i>-value
Intercept	2627.822	44.184	59.48	$< 2 \times 10^{-16}$
Age	-37.154	2.889	-12.86	1.64×10^{-10}

$$R^2 = 0.9018 \quad \text{Residual Standard Error} = 96.11 \text{ on } df = 18$$

Figures and tables must be numbered and presented with a caption. See Figure 1 and Table 1 in the template. When referring to a figure or a table, do not write “The figure below displays ...”, you should write “Figure 1 displays ...”.

Explain the results in full sentences, no point form.

For a log-linear model

$$\log(y) = \beta_0 + \beta_1 x + \epsilon,$$

exponentiate the coefficient gives the multiplicative factor for every one-unit increase in the regressor variable. For example, $\hat{\beta}_1 = 0.198$, $e^{0.198} = 1.218962$. For every one-unit increase in x , y is estimated to increase by a factor of about 1.22, or 22%. Similarly, if $e^{\hat{\beta}_1} = 0.84$, for every one-unit increase in x , y is estimated to decrease by a factor of $1 - 0.84 = 0.16$, or 16%.

9. Section 3: Conclusions and Recommendations

Report the results in APA style. A sample is given below.

A simple linear regression was calculated to predict shear strength of a propellant based on its age. A significant regression equation was found ($t(18) = -12.86$, $p = 1.64 \times 10^{-10}$), with $R^2 = 0.9018$. The predicted shear strength of a propellant is equal to $2627.822 - 37.154 (\text{Age})$ psi when age is measured in weeks. As the age of the propellant increases by one week, its shear strength is estimated to decrease by 37.154 psi.

Discuss the strengths and weaknesses of the results and give recommendations to overcome the weaknesses. Be specific to this study.

10. Section 4: R Code and R Output

Attached all the R code and R output used in this project.

Use `verbatim` to attach the code and output. Use `includegraphics` to attach figures. A few lines are shown in the template.

Every output must be preceded by a code, on the same page. Output that comes as a group should also be placed on the same page.

11. Proofread your work and check for consistency.

MARKING RUBRICS

Component Title	Report					Total Marks	100
Criteria	Score and Descriptors					Marks	
	Excellent	Good	Average	Need Improvement	Poor		
Introduction	10	8	6	4	2		
Data Analysis - exploratory	10	8	6	4	2		
Data Analysis - formal test	10	8	6	4	2		
Data Analysis - validation	10	8	6	4	2		
Data Analysis - predictions	10	8	6	4	2		
Conclusions	10	8	6	4	2		
Recommendations	10	8	6	4	2		
R Code and R Output	10	8	6	4	2		
Use of language is appropriate, with no grammatical errors.	10	8	6	4	2		
Overall work is neat, well-organised and creative.	10	8	6	4	2		
TOTAL							